



SKORE: Skill-oriented extraction engine from job advertisements using large language models

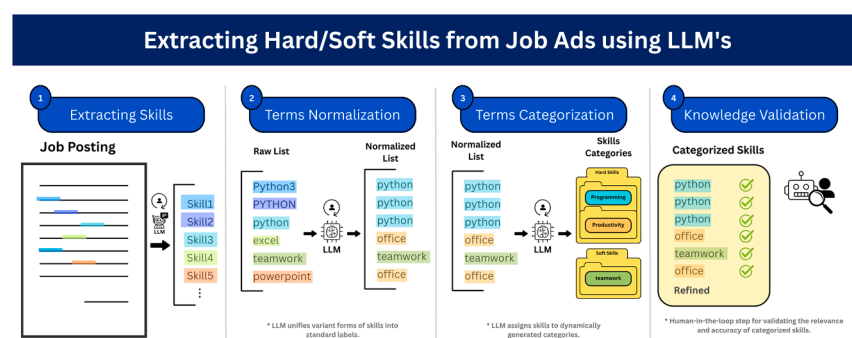
Pedro C. Menezes^{a,*}, Kenzo M. Sakiyama^b, Ricardo M. Marcacini^b,
Antonio F.L. Jacob Jr.^c, Fábio M.F. Lobato^{a,b,c,*} 

^a UFOPA, Federal University of Western Pará, Brazil

^b USP, University of São Paulo, Brazil

^c UEMA, State University of Maranhão, Brazil

GRAPHICAL ABSTRACT



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ABSTRACT

The digital transformation of the labor market has increased the number of Online Job Advertisements, which offer valuable data on job requirements and skills. However, extracting skills from this data is challenging because it is unstructured and lacks standardized analysis. Lexicon-based and supervised methods use predefined, labeled skill sets, which limit adaptability. Unsupervised methods often lack robustness and semantic clarity. Large Language Models (LLMs) have opened new possibilities for text analysis, but their use is often ad hoc. To address this, we present the Skill-Oriented Extraction Engine (SKORE), a framework for extracting, normalizing, and categorizing skills from Online Job Advertisements. It combines the semantic power of LLMs with human-in-the-loop approaches. The framework is model-agnostic to support a wide range of

* Corresponding authors.

E-mail addresses: pedro.menezes@discente.ufopa.edu.br (P.C. Menezes), fabio.lobato@ufopa.edu.br (F.M.F. Lobato).

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domains and data sources. A proof-of-concept case study shows that it can capture both technical and behavioral skills from job-post data.

This framework provides information for professionals who wish to enter a particular domain or transition careers.

SKORE also assists in creating Online Job Advertisements that reflect the real needs of the market.

Its applicability is envisioned for educational institutions (e.g., technical schools and universities) to respond to the market's dynamic nature on an ongoing basis.

Specifications table

This table provides general information on your method.

Subject area	Computer Science
More specific subject area	Artificial Intelligence, Job Post Analysis
Name of your method	Skill-Oriented Extraction Engine (SKORE)
Name and reference of the original method	N.A.
Resource availability	https://github.com/fabiolobato/SKORE

Background

The job market is constantly evolving, driven by technological changes and the increasing digitization of society [1]. This phenomenon has led to the emergence of Online Job Advertisements (OJAs), which represent unstructured information about job openings on the web. These ads usually include details such as job responsibilities, salary range, and required skills [2]. OJAs are a valuable source of information, as they reflect market needs and enable monitoring of trends in a given sector [3]. Mapping the set of necessary skills can benefit three actors: individuals seeking to enter or reposition themselves in the market; employers who aim to identify profiles of professionals who are more aligned with their needs while also analyzing their competitors; and education and training institutions, which adjust their curriculum guidelines to be more aligned with market needs [4].

Despite the vast knowledge contained in OJAs, automatically extracting skills remains a significant challenge, given their unstructured nature and the large volume of information available [5]. Currently, the main approaches in the literature are the

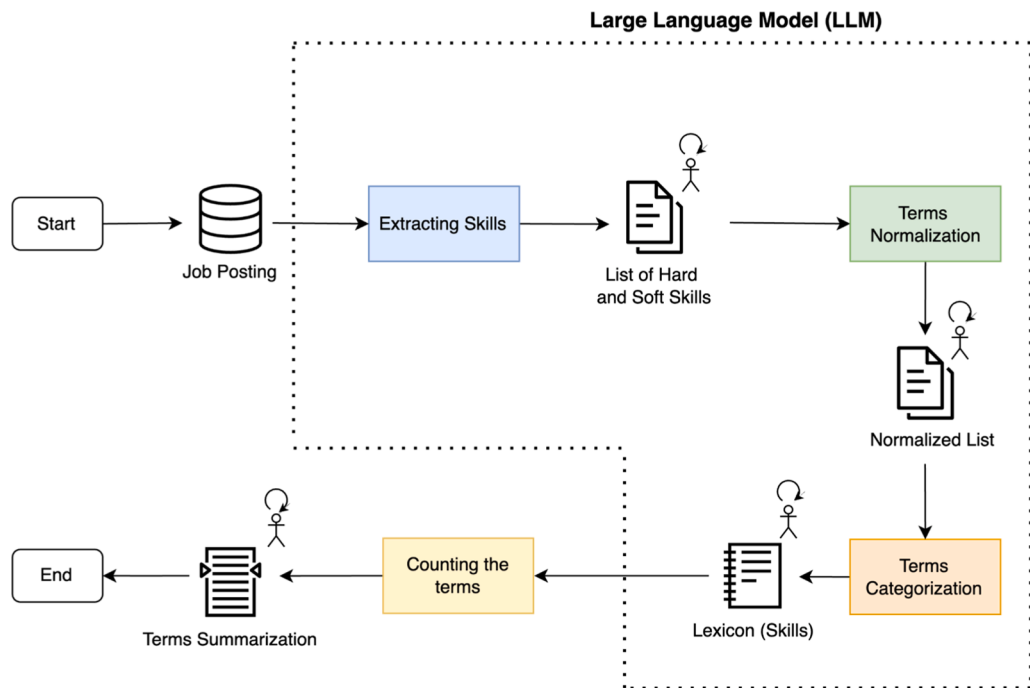


Fig. 1. The SKORE working flow.

Taxonomic Skills Dictionary [6] and Machine Learning (ML) algorithms [7]. Traditional Lexicon-based methods or Supervised ML approaches tend to rely on predefined sets of terms or manually annotated datasets, which limits their adaptability to new skills and their generalization across different sectors [8,10]. Moreover, they require manual annotation, which is labor-intensive, time-consuming, and error-prone [9].

On another spectrum, current unsupervised ML approaches face challenges with low accuracy and robustness, including the need to follow human-in-the-loop principles, sensitivity to preprocessing, and a lack of in-depth semantic interpretation [11]. With the advent of Large Language Models (LLMs), it is now possible to use them to identify skills and competencies in job advertisements. However, there are some limitations, such as extremely specific contexts that cause a degradation in response and possible model hallucination [10], or new technologies that do not yet exist at the time of model training and may be neglected (out-of-vocabulary words).

Therefore, there is a clear need for structured, reproducible approaches to extract knowledge from OJAs using LLMs while maintaining the reliability of the results. The proposed approach, called Skill-Oriented Extraction Engine (SKORE), addresses this gap by combining the potential of LLMs and incorporating domain expert supervision, adhering to the human-in-the-loop principle. The framework organizes the extraction process into well-defined stages, enabling more consistent and adaptable analysis of skill-related information. The outputs can aim to provide information for professionals who wish to enter or relocate in the job market, while also contributing to the creation of OJAs that meet companies' real needs. Moreover, it enables educational and training institutions to continuously respond to the market's dynamic nature. The contribution is not limited to the analysis of OJAs, which served as proof of concept. Therefore, it can be extended to other application domains such as health, business, and education.

Method details

The SKORE framework identifies skills in OJAs by combining text mining, Natural Language Processing (NLP), *prompt-tasking*, and LLM techniques. The Python programming language was used for design and implementation, while Deepseek-V3 was used because it presented a good cost-benefit ratio between value and performance in the tests for extracting aspects [12]. However, it is essential to note that SKORE is model-agnostic and can be adapted for use with other LLMs and programming languages. The framework was divided into four stages, as shown in Fig. 1: i) Extracting Skills (Hard and Soft); ii) Terms Normalization; iii) Terms Categorization; iv) Counting the Terms. At the end of the steps, the results are evaluated by humans, following the principles of the human-in-the-loop approach, to ensure the reliability of the results. Each step is described in the following subsections.

Extracting (Hard and soft) skills

Hard skills refer to technical knowledge and abilities acquired through education, training, and professional experience, while soft skills correspond to behavioral and interpersonal skills, such as communication, leadership, and teamwork [13]. The first step is to extract both hard and soft skills from the job ad descriptions. To do this, a structured dataset containing these descriptions is used. The

Table 1
Prompt structure for skill extraction.

Prompt	You are an assistant that extracts relevant skills from job descriptions. Your goal is to identify and return all hard and soft skills mentioned in the text. Classify each skill as either: - Hard skill: technical abilities, tools, programming languages, or methods. - Soft skill: interpersonal, emotional, behavioral, or cognitive traits. Only include skills that are explicitly mentioned in the job description. At the end, include a "reasoning" field: a short paragraph (2-3 sentences) explaining the reasoning behind the overall result. If no skills were found, explain clearly why (e.g., no explicit skills mentioned, description too vague, etc.). Return the output in the following JSON format: { "hard_skills": ["skill1", "skill2", ..], "soft_skills": ["skill1", "skill2", ..], "reasoning": "Your overall explanation here." } Given the following job description, extract all hard skills and soft skills mentioned. Return the output as a JSON object with two keys: "hard_skills" and "soft_skills". Do not infer skills not explicitly mentioned. Job Description: "" [JOB DESCRIPTION] ""
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process is iterative for each job ad, using the prompt-tasking technique to identify skills, along with a brief explanation for each result.

Output Example	<pre>{ "hard_skills": ["decision analysis process", "probability distributions", "decision tree", "influence diagrams"], "soft_skills": ["Communication", "Coordination", "Problem-solving", "Proactive measures implementation"], "reasoning": "The hard skills are explicitly mentioned in the job description, focusing on technical areas like Natural Language Processing, Explainable AI, and specific machine learning techniques. The soft skills are derived from the description's requirements for interpersonal abilities, teamwork, communication, and resilience under pressure, all of which are clearly stated as necessary for the role." }</pre>
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The *prompt* was constructed in accordance with the best practices of Prompt Engineering, clearly describing the task to be performed, the context, the input data, and the expected output format. Table 1 illustrates the structure of the *prompt* used and provides an example of generic output. The prompt design was built using the Zero-shot technique, given the difficulty of accessing datasets with skill-labeled descriptions, a challenge that can be explored in future research [14]. To avoid possible hallucinations, the prompt was treated as an extraction and classification problem divided into stages, in which the model must:

1. Extract skills;
2. Classify each skill as *hard* or *soft*;
3. Exclude inferred skills that are not listed in the Job Ad (possible hallucination);
4. Generate the response in JSON format.

As mentioned earlier, this step involves an iterative process in which the space reserved for [JOB DESCRIPTION] – as shown in the prompt (Table 1) – is updated with each iteration to reflect the textual description of the respective Job Ad. To standardize the output,

Table 2

Prompt structure for the Terms Normalization step.

Prompt	<pre># System Message: Skill Normalization Expert You are an expert in **Natural Language Processing (NLP)**, specializing in **text normalization and ontology mapping for job-related skills**. Your task is to process a list of job-related skills and return a structured dictionary, grouping variations of the same skill under a standardized term. ## **Normalization Rules** 1. **Identify and group similar skills** that have spelling variations, plural/singular forms, or slight semantic differences. 2. **Use the most common industry-standard term** as the primary key. 3. **Exclude skills that do not have any variations**—if a skill does not have at least one synonym or alternative form, it should not be included in the final dictionary. 4. **Merge synonyms and abbreviations** under the same key (e.g., "CRM Management" should include "CRM Systems", "CRM Software Management", and "CRM"). 5. **Preserve distinctions between different concepts** (e.g., "Marketing Strategy" vs. "Content Marketing Strategy"). 6. **Group soft skills logically**, avoiding redundant listings (e.g., "Strategic Thinking" should include "Big-Picture Planning", "Visionary Thinking"). 7. **Group tools under their parent suite**: - If a tool is part of a larger software suite, group it under the suite name. - Example: "Excel", "Word", and "PowerPoint" should be grouped under "Microsoft Office". 8. **Group related concepts under broader categories**: - If terms represent similar concepts, they should be grouped under a more general category. - Example: "data visualization", "predictive modeling", and "segmentation analysis" → "data analysis". 9. **Handle common abbreviations and stylistic differences**: - Group terms that are commonly used interchangeably in the industry. ## **Expected Output Format** A valid **Python dictionary (JSON)** where: - **Keys** are the **standardized skill names**. - **Values** are **lists of alternative names, abbreviations, and spelling variations**. - **Entries with no variations should be excluded from the dictionary** to optimize token usage. ## **Input** [SKILL LIST] { "Python Programming": ["Py", "Python Dev", "Python Language"], "Microsoft Office": ["Excel", "Word", "MS Office", "PowerPoint"], "Data Analysis": ["Analytics", "Data Modeling", "BI Analysis"] }</pre>
Output Example	<pre>{ "Python Programming": ["Py", "Python Dev", "Python Language"], "Microsoft Office": ["Excel", "Word", "MS Office", "PowerPoint"], "Data Analysis": ["Analytics", "Data Modeling", "BI Analysis"] }</pre>

the JSON format is explicitly described with three components: (i) a list of hard skills, (ii) a list of soft skills, and (iii) a reasoning field containing the logical justification for the extraction performed. If the model identifies that the description does not correspond to a Job Ad (e.g., institutional descriptions or generic requirements), the lists are returned empty, and the “reasoning” parameter describes the reasoning for that response. At the end of the iterative process, two lists of unique skills are constructed, one for hard skills and one for soft skills, by merging the results of each iteration. These artifacts will serve as the basis for the next step.

Terms normalization

Term normalization consists of the lexical standardization process, in which morphological, orthographic, or synonymous variants of the extracted skills are mapped to a common term to ensure consistency in the results [15]. This step is critical because there is a high degree of variation in skill expression during the extraction process. For example, the terms “Python” and “Python Programming Language” are different but have the same meaning. Thus, the second step aims to reduce variation among similar terms.

SKORE performs normalization using *Prompt Engineering*, a *prompt-tasking* approach that aims to overcome the limitations of traditional normalization techniques, since LLMs can capture semantic nuances. The *prompt* used is described in Table 2, which instructs the model to act as an NLP expert by following nine rules for normalization. This design incorporates terminology hierarchization, preservation of conceptual distinctions, and token optimization.

At the end of the process, two normalization dictionaries are obtained, one for hard skills and the other for soft skills. In both, the keys represent the normalized term, and the values are lists of skills associated with each key. The generated dictionaries are submitted for human evaluation to analyze the consistency of the LLM results. Finally, after expert validation, the dictionaries are applied to the lists from the first stage, replacing the necessary terms with their respective keys, yielding two normalized skill sets (hard and soft).

Table 3
Prompt structure for the first subtask: Terms Categorization step.

Output Example	Prompt Your task is to analyze a large list of [HARD - SOFT] skills of [DOMAIN] commonly found in professional environments, especially in the fields of technology, data, engineering, and design. You MUST create a set of **[NUMBER OF CATEGORIES] categories** that group these skills based on: - Semantic similarity - Domain of application (e.g., software development, data science, design, infrastructure, etc.) For each category, provide: - A **clear title** (e.g., "Web Development") Instructions: - Use **semantic similarity** and **domain knowledge** to group skills into meaningful, practical categories. - Each category should represent a distinct area of expertise or type of technical competency. - Keep category names short and commonly used in the industry. - Return **only the list**, starting with: NONE [Category Name] [Category Name] [Category Name] .. Here is the full list of hard skills: [LIST OF STANDARDIZED SKILLS] Use this full set to define the categories. Please generate the list of categories as described.
	Output NONE Data Management & Analytics Software Development & IT Systems Supply Chain & Logistics Management Financial Management & Analysis Information Security & Compliance Healthcare Informatics & Management Manufacturing & Production Operations Process & Project Management Decision Science & Optimization Emerging Technologies (AI, IoT, Blockchain) Compliance & Regulatory Affairs Transportation & Fleet Operations Engineering & Technical Systems Administrative & Clerical Support Risk Management & Quality Assurance

Terms categorization

Term categorization consists of grouping sets of semantically equivalent terms [16]. This stage aims to reduce dimensionality and provide more meaningful results. The process was divided into two subtasks: i) the creation of categories and ii) the individual classification of skills. For the first subtask, the *prompt* described in Table 3 was used, which has as inputs the number of categories, the application domain, and the normalized list of skills. Additionally, the *prompt* defines rules that consider semantic similarity and the output format, adhering to best practices in *Prompt Engineering*. These characteristics enable the model to create categories based on the lexical diversity of the skills. Furthermore, it is worth noting that for each skill class (hard and soft), its respective categories were generated.

The second subtask involves classifying skills according to previously defined categories. As illustrated in Table 4, the corresponding *prompt* aims to map each skill, whether hard or soft, to the most appropriate category, while also considering criteria such as context, domain, and meaning. The classification process is conducted interactively, with each skill processed individually within the established set of categories. If the model is unable to map the category, the term is assigned to “*none*.” It is crucial to emphasize that this procedure is applied separately to the hard-skill and soft-skill lists, as they are distinct sets of skills. This strategy ensures atomicity of operations and consistency in category assignment. At the end of this stage, the lexicon artifact is obtained, containing the skills divided into types and categories.

Counting terms

The final stage of term counting aims to quantify the relative frequency of the skills and categories identified in the previous stages. By counting term frequency, SKORE can reveal patterns and competencies that stand out in the context. This enables the highlighting of key competencies and mapping of emerging trends in the analyzed sector. To facilitate the interpretation of the results, this stage involves constructing a visual summary of the data using graphs that synthesize the information in an accessible manner. These representations facilitate the validation of results by human experts, assist in delivering knowledge, and provide a more objective analysis of market demand.

Table 4
Prompt structure for the second subtask of the Terms Categorization stage.

Prompt	You are a semantic classification engine trained to map individual [HARD - SOFT] skills to their most appropriate category. Your task is to assign one single category to each skill, using semantic similarity and domain relevance. You must use the list of categories provided earlier. When choosing the category, consider: - The skill's technical context - The domain of application - Its semantic meaning If a skill could belong to more than one category, pick the most specific and relevant one based on common industry usage. Instructions: - Use the categories provided by the user. - You MUST return the output in the exact format shown: - [Skill] → [Category Name] No explanation, the output needs to be only skills and the category Using the following categories, assign each skill below to one category only. Categories: <pre>""" [CATEGORIES] .. """</pre> Skill to categorize: [SKILL] Instructions: - Use the categories provided by the user. - You MUST return the output in the exact format shown: - [Skill] → [Category Name] The output needs to be in that format programming tools (visual basic and excel macros) → software development & it systems
Output Example	

Method validation

SKORE is a framework for extracting skills from job ad descriptions by integrating NLP, LLMs, prompt-tasking, and human-in-the-loop techniques, making it easily adaptable across contexts. As a proof of concept, a brief case study was developed in the field of Operational Research (OR), following the guidelines of the research method known as Design Science Trajectories (DST) [17]. DST is an evolution of the traditional and well-established CRoss-Industry Standard Process for Data Mining (CRISP-DM), covering more exploratory nuances of the data and allowing researchers to choose the research flow [18,19]. As the objective of the case study is to demonstrate SKORE's practical applicability and validate its performance in real-world scenarios, three stages were selected, as shown in Fig. 2: i) Data Acquisition, ii) Modeling, and iii) Results Exploration.

Data acquisition

The data were obtained through web scraping of OJAs. The keywords related to the domain under scrutiny were defined using the Delphi methodology, drawing on terms from renowned journals in the field, such as the Operations Research Journal (OPRE) and the European Journal of Operational Research (EJOR). With the terms initially mapped in the “*aims and scope*” and “*topics of interest*,” the authors discussed their relevance among themselves and reached a consensus. Pilot searches were conducted to assess the noise of each term used, for example, the term “returning vacancies” was not related to Operational Research. In the end, the following keywords were used in data collection: *Computational Intelligence; Decision Analysis; Decision Support; Discrete Optimization; Information Management; Manufacturing Logistics; Operations and Supply Chain; Production Logistics*. In the end, 413 OJAs were collected from platforms such as seek.com, jobs.lever, and Times Higher Education.

Modeling

Following the processes and configurations described in the **Method Details** section, we arrive at the modeling stage. The Deepseek-V3 model was used because it presented a good cost-benefit ratio between value and performance in the aspect extraction tests. The process followed the steps outlined in Fig. 1 and their corresponding prompts described in Tables 1–4. Implementation details are available in the GitHub repository listed in the Specifications table.

Results exploration

The implementation of the SKORE yielded a total of 3470 skills, comprising 2371 unique hard skills and 1099 unique soft skills. After normalization, 2113 and 784 were obtained for hard and soft skills, respectively, representing a reduction of approximately 17 %. This is a conservative approach, and it is possible to reduce it further, potentially at the expense of information about the vacancies. These values show that although the dataset is considered small, the information in the job descriptions has high information value for analyzing market demand. Regarding the disparity between hard and soft skills, the greater diversity of hard skills reflects both the specificity required and the domain's lexical diversity, since the list of unique hard skills is more than double that of soft skills.

Fig. 3 shows the most frequent skills after data normalization. Among hard skills, those such as “programming language,” “data analysis,” and “Microsoft Office” stand out as essential in contexts involving mathematical modeling, computer simulation, and logistics optimization — the foundations of Operations Research. Competencies such as “warehouse management,” “transportation,” and “forklift operation” may highlight adherence to operational management and supply chain practices. The presence of “artificial intelligence” and “data visualization” reinforces the framework's ability to identify emerging competencies, demonstrating its potential to capture current trends in the field. Regarding soft skills, teamwork, attention to detail, and leadership skills are the most prevalent, as they relate to the demands of professionals who work in complex problem-solving and team management. The terms “communication skills” and “adaptability” complement the professional profile, both of which are relevant in organizational contexts.

Fig. 4 represents the distribution of categories constructed by the model. Regarding hard skills, categories such as “software development & systems” stand out, followed by “supply chain & logistics management,” “data analytics & visualization,” and “transportation & fleet operations,” indicating the relevance of these skills in developing decision support applications, for example. The terms also reveal the centrality of topics such as logistics, data analysis, and transportation operations, corroborating the results presented in Fig. 3. Skills such as “cybersecurity & information security” and “engineering & systems integration,” which, although less frequent, demonstrate that the proposed framework is capable of capturing specific niches of required skills. It is also worth mentioning the “none” category, which represents skills that do not fit into any category, guiding refinement and category expansion.

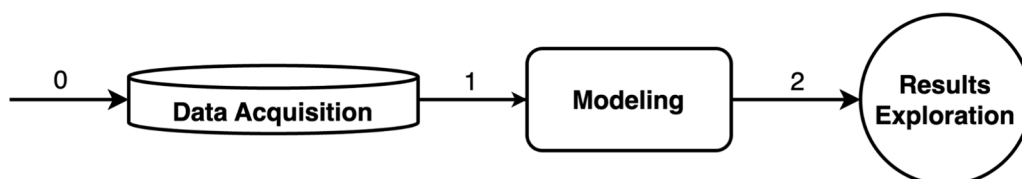


Fig. 2. DST flow.

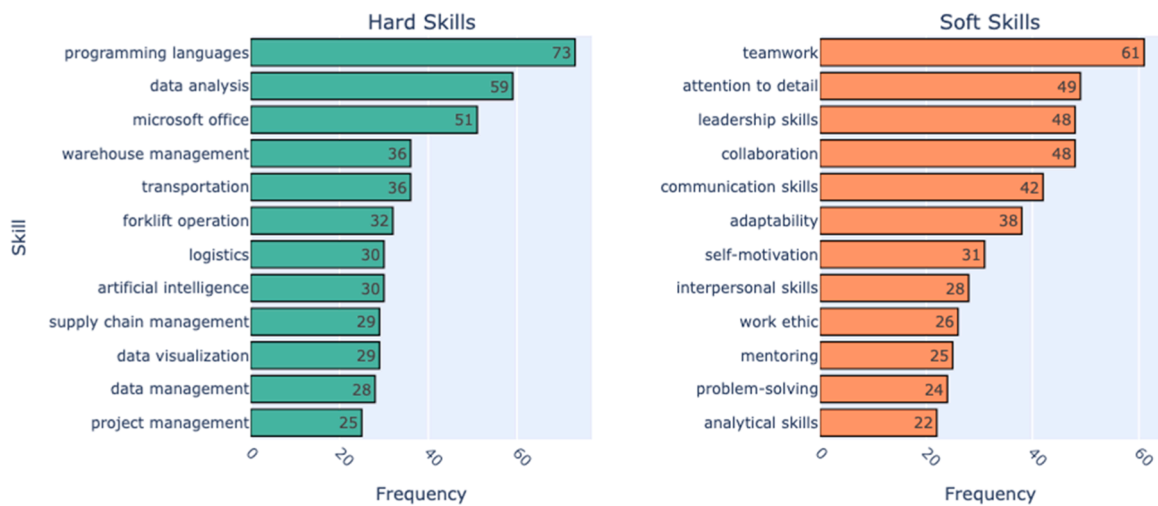


Fig. 3. Twelve most frequent skills (Standardized List).

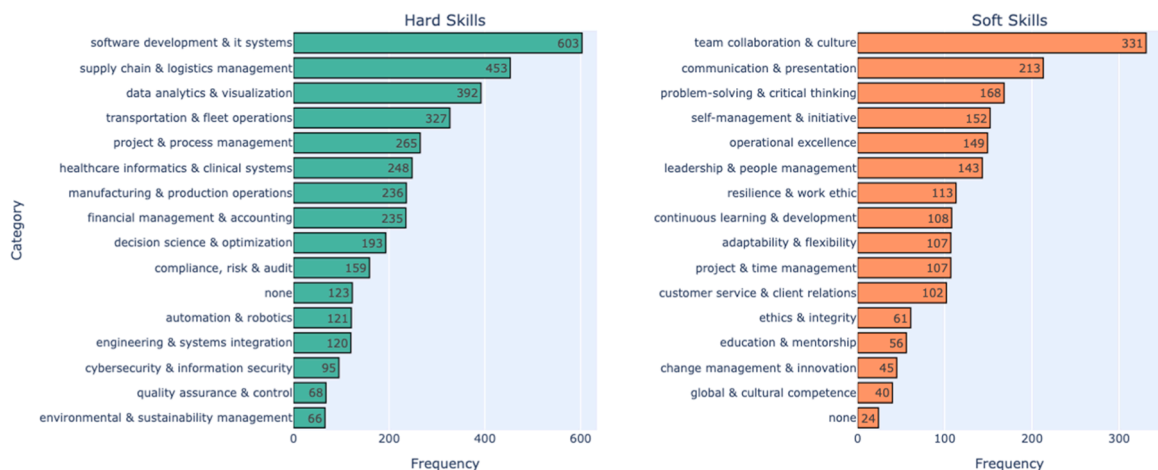


Fig. 4. Frequency of hard and soft skills categories extracted.

Among soft skills, the most frequent categories were “team collaboration & culture,” “communication & presentation,” and “self-management & initiative.” These skills are known to be essential in projects involving collaboration, problem-solving, and team leadership, common conditions in the OR field. Other categories, such as “adaptability & time management”, and “ethics & work ethic”, confirm the relevance of adaptive and ethical capabilities in solving organizational challenges.

Final remarks

Extracting skills from OJAs presents a persistent challenge due to the dynamism of the job market. Traditional Lexicon-based and Supervised Learning approaches require robustly annotated datasets, which prevents them from generalizing to different sectors. In the current literature, there is a lack of robust, widely recognized methodologies that effectively address this problem, thereby imposing significant obstacles to direct performance comparisons.

To overcome this methodological limitation, this study highlights SKORE, a framework developed as an agnostic approach to automating skill extraction in OJAs using Large Language Models (LLMs). The framework is structured in four sequential stages: 1) skill extraction; 2) term normalization; 3) term categorization; and 4) term counting. The feasibility and potential impact were demonstrated through experiments described in the *Results Exploration* subsection and validated by domain experts, confirming that the approach achieves the proposed objectives and begins to address this research gap.

SKORE's central contribution lies in proposing a scalable, adaptable solution that adapts to the dynamics of the labor market. It proves valuable for analyzing sector trends and mapping both emerging and latent skills across various contexts. For future work, we intend to explore adapting the SKORE to apply fine-tuning techniques to optimize performance in specific domains. In addition, future research could expand the validation to a larger dataset and explore integrating SKORE with other sources of information for skill

analysis. Additionally, the framework can be leveraged to annotate structured datasets, enabling the creation of labeled corpora for training tasks. In this context, SKORE may also be used within teacher–student frameworks (e.g., knowledge distillation), where its outputs serve as supervisory signals for training smaller, more efficient models [21].

Limitations

SKORE proposes using an LLM to assist in extracting skills from OJAs. Although innovative, the framework has limitations in its use of Artificial Intelligence (AI) to solve practical problems, which are briefly described below. *Limitations associated with LLM*: One of the main problems faced by LLM-based systems is the potential for hallucinations, whereby the model generates erroneous or incoherent results [20]. In this sense, the model can interpret the prompt incorrectly and, for example, add skills that are not in the job description. To address this, instructions that follow Prompt Engineering best practices were defined at all steps of the framework, aiming to mitigate part of the hallucination process. *Stochastic inconsistency between executions*: LLMs operate based on probabilistic text generation models, in which each subsequent token is determined by a learned probability distribution. LLMs may yield non-deterministic outputs for identical prompts, which poses a challenge in contexts where reproducibility is required. The model can generate different results for the same input and select different terms in different rounds.

To address the limitations of LLMs, measures were implemented to ensure consistent results. Firstly, to mitigate the occurrence of hallucinations, it is essential to provide clear instructions, as documented in the **Method Details** section. This reduces the occurrence of incoherent results and yields more concise results for skill extraction. To help reduce inconsistency between rounds, the best parameters were defined to ensure that the model is deterministic (e.g., temperature=0). In addition, it is worth highlighting the importance of human-in-the-loop approaches, which enable human interaction in the modeling process, thereby increasing the reliability of the results.

CRedit author statement

Pedro C. Menezes: Conceptualization, Methodology, Validation, Formal analysis, Investigation, Writing – original draft, Writing – review & editing. **Kenzo M. Sakiyama**: Validation, Formal analysis, Writing – review & editing. **Ricardo M. Marcacini**: Conceptualization, Methodology, Validation, Writing – review & editing. **Antonio F. L. Jacob Junior**: Conceptualization, Methodology, Validation, Writing – review & editing. **Fábio M. Lobato**: Conceptualization, Methodology, Validation, Formal analysis, Investigation, Writing – original draft.

Related research article

None.

For a published article

None.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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